

Technical Appendix for SPRiF: Spectral Projective Reset Integrate-and-Fire Neuron

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1 Supplementary Guide and Evidence Map

This technical appendix expands the evidence used in the main paper rather than introducing a second line of claims. It provides reproducibility details for the five benchmark results, exact definitions for the mechanism variants and SI-DMS intervention, and the formal SPRiF update plus dynamical-analysis protocols.

Consistency locks. All sections preserve the main-paper contract: the slow spectral state is three-dimensional, the fast discharge state is two-dimensional, and each timestep follows $I_t \rightarrow \mathbf{x}_t \rightarrow \tilde{\mathbf{u}}_t \rightarrow v_t \rightarrow z_t \rightarrow \mathbf{u}_t$. Projective reset acts directly only on the fast discharge state, with $\Delta \mathbf{x}_t^{\text{reset}} = \mathbf{0}$; in recurrent layers, spikes may still affect the next slow-state update through the ordinary $W_{\text{rec}} \mathbf{z}_t$ input pathway. We use the main-paper terms *functional state decomposition*, *slow spectral state*, *fast discharge state*, and *projective reset* without redefining them. Benchmark SPRiF entries are five-seed mean $\pm$ standard deviation, SI-DMS entries are three-seed summaries, and cross-paper comparator entries remain descriptive comparisons.

Evidence map. Each section below expands a specific main-paper element rather than introducing a second line of claims:

- Main-paper Table 1 is expanded by the dataset encodings, architectures, optimization settings, and reporting rules in Sections ??–??. The SPRiF entries remain five-seed estimates and published comparator entries remain descriptive context.
- Main-paper Table 2 is expanded by the variant definitions in Section ??: A is $\omega = 0$, B is the merged slow/fast state that changes both state role and total state dimension, and C is $\lambda = 0$.
- The SI-DMS results are expanded by the recurrent task construction, paired-loss definition,

intervention-fraction sweep, delay-by- K coverage, and integrity audit in Sections ??–??. Training uses $K \in \{0, 1, 2, 4, 8\}$ at a fixed 15% intervention fraction; evaluation extends to $K \in \{16, 32, 40\}$ and fractions from 5% to 50%.

- The S-MNIST loss-landscape diagnostic in Section ?? compares one saved SPRiF checkpoint with one saved LIF checkpoint on a shared perturbation-coordinate grid. Because it uses one training seed and one random direction pair, it is treated as a checkpoint-level optimization diagnostic rather than a multi-seed generalization result.
- The mechanism and temporal-kernel figures are expanded by the reset identities and impulse/spectral protocols in Sections ??–??.
- A fixed-checkpoint feature-perturbation diagnostic is reported separately in Section ??; it records how saved checkpoints respond under feature perturbations without adding a new benchmark, robustness ranking, or biological interpretation for λ .

2 Experimental Protocols and Reporting

The five benchmark entries use task-standard inputs and readouts. S-MNIST and PS-MNIST use MNIST ? tensors read as 784-step pixel sequences; PS-MNIST applies a seed-generated random pixel permutation fixed within the run. QTDB uses the public QT Database ? and adopts the preprocessing protocol from *Accurate and Efficient Time-Domain Classification with Adaptive Spiking Recurrent Neural Networks* ?: two normalized ECG channels are converted by increasing and decreasing level-crossing events into four input event streams, and a time-distributed classifier predicts six waveform classes at each timestep. The released QTDB utilities expose `convert_seq`, which applies Savitzky–Golay smoothing followed by thresholded positive/negative

first-difference events, and `convert_dataset_wtime`, which unpacks the encoded `x`, `y`, and `t` arrays and derives inter-timestep intervals. Training consumes the resulting `QTDB_train.mat` and `QTDB_test.mat` files. GSC uses one-second utterances from Google Speech Commands ?, represented as 40-bin mel features with first- and second-order deltas stacked over 101 frames, followed by temporal mean pooling over hidden spikes. SHD uses the 700-channel cochlea event sequences from the Spiking Heidelberg Digits dataset ?, with recurrent hidden processing and accumulated readout evidence after warm-up.

The benchmark set was selected to cover complementary temporal regimes rather than to form a single matched leaderboard. S-MNIST tests retention over a long serialized image, PS-MNIST removes spatial locality through a fixed permutation, QTDB requires dense time-distributed physiological-stream decisions, GSC tests dense acoustic features over an utterance, and SHD tests sparse event-driven input. Together they span sequence structure, modality, and readout conventions; comparisons remain task-specific because the training protocols are not identical across datasets.

2.1 Architectures and Optimization

Table ?? reports the task-specific network and optimization settings for the submitted SPRiF runs. Ranges for ω are expressed as multiples of π .

For S-MNIST and PS-MNIST, truncated BPTT uses a chunk length of 262 steps. The GSC network uses temporal mean pooling over hidden spikes before the linear classifier. The QTDB network optimizes the time-distributed sequence loss. SHD accumulates readout evidence after the warm-up period. All models use the multi-Gaussian surrogate gradient described in Section ??.

During development, candidate settings were explored within the following envelopes: learning rate $[4 \times 10^{-5}, 5 \times 10^{-2}]$, threshold $[0.3, 1.5]$, $\tau_\alpha \in [1, 140]$ steps, $\tau_\rho \in [1, 80]$ steps, $\tau_\eta \in [0.2, 20]$ steps, and $\omega \in [0.1, 1.0]$ radians. Final task-specific settings were selected by validation accuracy and then fixed across the five reported seeds. These envelopes describe the limits used during development, not an exhaustive Cartesian grid; Table ?? reports the final settings and initialization intervals used for the submitted runs.

2.2 Seeds, Model Selection, and Reporting

All five benchmarks use the seed set $\{0, 1, 2, 42, 1111\}$. For each benchmark and seed, the checkpoint is se-

lected according to validation accuracy, and Table 1 reports the mean and standard deviation of the resulting evaluation accuracies over the five independently seeded runs. Within a task, all runs use the same task-specific preprocessing, data splits, and architecture settings in Table ??; only the seeded training realization changes. Published comparator entries provide cross-paper context rather than matched controls. SI-DMS is a separate recurrent mechanism experiment using seeds $\{1, 2, 3\}$. For its summary statistics, each seed is first averaged over the five evaluation delays; the reported mean and sample standard deviation are then computed over the three seed-level values.

The released benchmark utilities seed Python’s `random` module, NumPy, PyTorch on CPU, and all visible CUDA devices; they also disable cuDNN benchmarking. SI-DMS applies the same four random-number seeds and additionally constructs condition-specific `torch.Generator` instances so that compared models receive identical cue tensors and intervention masks within each seed/delay/ K /fraction condition. These controls reproduce the principal random choices, although exact bitwise identity across different hardware and software stacks is not claimed.

2.3 Evaluation Metrics and Uncertainty

Accuracy is the primary metric because every benchmark is a single-label classification task and the cited comparator literature reports accuracy. S-MNIST, PS-MNIST, GSC, and SHD use the fraction of correctly classified sequences; QTDB uses the fraction of correctly classified labeled timesteps under its time-distributed readout. SI-DMS uses sequence-level match/non-match accuracy together with the paired intervention loss in Eq. (??). Accuracy measures task capability, while the paired loss separates the additional degradation caused by controlled spike interventions from differences in clean accuracy.

For S independent training seeds with seed-level accuracies a_s , we report

$$\bar{a} = \frac{1}{S} \sum_{s=1}^S a_s, \quad s_a = \sqrt{\frac{1}{S-1} \sum_{s=1}^S (a_s - \bar{a})^2}. \quad (1)$$

Using the displayed means and standard deviations, the five-seed SPRiF entries in the main benchmark table approximately correspond to the following two-sided 95% t intervals for the seed mean: S-MNIST [99.21, 99.35], PS-MNIST [95.69, 96.03], QTDB [87.98, 88.88], GSC [94.17, 94.93], and SHD

Item	S-MNIST	PS-MNIST	QTDB	GSC	SHD
Input dim	1	1	4	120	700
Output dim	10	10	6	12	20
Layer	2 Rec (64, 256)	2 Rec (64, 256)	1 Rec (36)	1 Rec (300)	1 Rec (64) + readout
Epochs	150	150	250	150	100
Optimizer	Adam	Adam	Adam	AdamW	Adam
Batch	512	512	128	128	100
LR	1e-2	1e-2	1e-2	5e-3	1e-2
Loss	CE	CE	CE	CE	CE
LR decay	0.1 / 50 ep	0.1 / 50 ep	0.65 / 100 ep	0.5 / 30 ep	0.5 / 20 ep
LR decay type	StepLR	StepLR	StepLR	StepLR	StepLR
Threshold	1.0	1.0	0.6	0.8	1.0
τ_α (steps)	[2, 60]	[2, 60]	[20, 120]	[10, 80]	[10, 80]
τ_ρ (steps)	[2, 40]	[2, 40]	[4, 30]	[4, 30]	[4, 30]
τ_η (steps)	[0.8, 10]	[0.8, 10]	[0.8, 8]	[0.8, 8]	[0.8, 8]
ω ($\times \pi$)	[0.04, 0.50]	[0.04, 0.50]	[0.02, 0.20]	[0.04, 0.40]	[0.04, 0.40]

Table S1: Architecture and optimization settings for the SPRiF benchmark entries. “Rec” denotes a recurrent SPRiF layer, LR decay denotes the StepLR multiplier and interval, and CE denotes cross-entropy.

[90.95, 92.09] percent. These intervals quantify variation among the submitted SPRiF runs. They are not tests against published comparator values, whose seeds, data processing, architectures, and optimization protocols are not matched and whose per-seed results are generally unavailable.

2.4 Compute Environment

Experiments were run on one NVIDIA GeForce RTX 2080 Ti GPU with 11 GB of memory and 12 vCPUs of an Intel Xeon Platinum 8255C CPU at 2.50 GHz. The software environment used Ubuntu 22.04, Python 3.12, PyTorch 2.3.0, and CUDA 12.1. Training times range from minutes for QTDB to several hours for PS-MNIST and GSC, depending on sequence length and network width. No distributed training or specialized neuromorphic hardware was used.

2.5 S-MNIST Loss-Landscape Diagnostic

We probe local parameter sensitivity around trained S-MNIST solutions using a two-dimensional random-plane visualization inspired by filter-normalized loss landscapes ?. The SPRiF and LIF models use the same two recurrent layer widths, [64, 256], temporal-mean readout, normalized 784-step S-MNIST inputs, optimizer, learning-rate schedule, and seed. Their neuron dynamics differ, so the saved models contain 92,810 and 88,970 parameters, respectively; this is therefore an architecture-width-matched rather than parameter-count-matched comparison. The checkpoint metadata record test accuracies of 98.88% for

SPRiF and 95.19% for LIF. These values identify the analyzed single-seed checkpoints and do not replace the five-seed benchmark summaries in the main paper.

For each checkpoint θ^* , we generate two Gaussian random directions $\mathbf{d}_1$ and $\mathbf{d}_2$ using seeds 123 and 124. For every multidimensional parameter tensor W , its corresponding direction tensor D is rescaled as $D \leftarrow D\|W\|_F/(\|D\|_F + 10^{-10})$; one-dimensional parameters, including biases and neuron-wise dynamics parameters, are held fixed. We then evaluate

$$\mathcal{L}(a, b) = \mathcal{L}(\theta^* + a\mathbf{d}_1 + b\mathbf{d}_2), \quad (a, b) \in [-1, 1]^2, \quad (2)$$

on a 25×25 grid. Cross-entropy is estimated on the first four non-shuffled test mini-batches (2,048 images), with the 784-step sequence processed in chunks of 262 steps and the chunk-level logits averaged. Both models use the same coordinate range, grid resolution, and evaluation samples; the directions are normalized separately against each model’s checkpoint tensors.

Figure ?? shows a broader low-loss neighborhood for the analyzed SPRiF checkpoint. At the unperturbed center, the estimated cross-entropy is 0.0575 for SPRiF and 0.2098 for LIF. Over the 625 sampled coordinates, the SPRiF slice has lower loss at 85.1% of points, with mean losses of 10.69 versus 15.78 and maxima of 36.46 versus 50.85. Thus, under these norm-matched perturbations, this SPRiF solution is less sensitive along the sampled plane. The result is deliberately scoped to one checkpoint and one direction pair: it does not characterize the full high-dimensional landscape, isolate which SPRiF

component creates the geometry, or by itself establish a causal link between flatness and generalization.

3 Mechanism Variants and SI-DMS Evaluation

3.1 Mechanism Variant Definitions

Table 2 in the main paper reports the ablation outcomes. Table ?? defines the exact intervention behind each column without repeating the main-paper numbers.

SPRiF full, $\omega = 0$, and $\lambda = 0$ each retain three slow-state and two fast-state coordinates, for five local state values per neuron. The merged variant instead uses a single three-dimensional state. It therefore changes both functional separation and state capacity; its outcome diagnoses the combined consequence of merging the states and reducing the local state dimension, rather than a capacity-matched causal isolation of state separation alone.

3.2 SI-DMS Protocol

The Spike-Intervention Delayed Match-to-Sample (SI-DMS) task tests whether a recurrent network retains a cue through a delay while selected hidden units are forced to emit real spikes. SI-DMS is a programmatically generated mechanism task rather than a new static dataset: cue identities, Poisson spike realizations, delays, labels, and intervention masks are generated from the recorded seeds and condition settings. Table ?? gives the protocol and matched-control configuration.

For reproducibility of the main paper’s paired replay, we use the seed-1 SPRiF-full checkpoint and deterministic batch seed 0 at delay 1600 ms. Clean ($K = 0$) and stressed ($K = 8$, nominal 15% intervention; 10 of 64 hidden units per event) conditions receive the exact same A–A input tensor. The two A cues share stimulus identity but are independently sampled realizations of that identity. All 80 selected unit–timestep pairs in the stressed condition produce forced spikes (80/80 forced hits). Both conditions have match targets and predictions, and each cue-2 window contains 203 natural hidden-layer spikes across all 64 units. The raster uses one shared set of ten units, selected jointly after excluding units naturally active in more than 40% of delay steps. This pair is illustrative rather than a population statistic.

3.3 Condition-Level Grids

The condition-level result set covers six models, three seeds, five delays, eight intervention counts, and six positive-intervention fractions. Because clean accuracy is independent of intervention fraction, each model/seed file contains five clean rows and $5 \times 7 \times 6 = 210$ positive-intervention rows, for 215 rows per run and 3,870 rows in total. The main-paper figure selects the 15% fraction before aggregating; it never pools the six fractions into one K curve.

For seed s , delay d , count K , and fraction f , we define the paired loss

$$\Delta \text{Acc}_{s,d}(K, f) = \text{Acc}_{s,d}(0) - \text{Acc}_{s,d}(K, f). \quad (3)$$

Clean accuracy is retained as a model-capability measure. Equation (??) separately quantifies the additional loss caused by the intervention within the same trained run and evaluation delay. Table ?? reports the training-range boundary $K = 8$ and the unseen count $K = 40$ at $f = 15\%$.

We also report an unstandardized paired effect size at the unseen $K = 40$ condition. For each comparator, its seed-level intervention loss is paired with SPRiF full at the same seed after both are averaged over the same five delays. Relative to SPRiF full, the mean excess losses are 4.04 percentage points for $\lambda = 0$ (three-seed range 4.02–4.08), 36.19 for the merged variant (32.31–38.36), 31.03 for ASRNN (24.24–38.72), 25.38 for BRf (23.91–26.18), and 13.47 for LIF (2.40–22.50). The effect is most directly interpretable for $\lambda = 0$ and clean-matched ASRNN; merged, BRf, and especially LIF have different clean capability or seed variability. With only three independent seeds, an exact two-sided sign-flip test has minimum attainable $p = 0.25$, so we do not make a binary significance claim or treat the five delays as independent replicates.

Figure ?? holds $K = 40$ fixed and varies the fraction of hidden neurons forced to spike at each event. At 20%, SPRiF full and $\lambda = 0$ have accuracies of 99.2% and 94.2%, respectively, whereas ASRNN has 58.1%; the merged variant is near chance. BRf and LIF are shown together with their lower clean accuracies rather than normalized to a common starting point.

3.4 Model Controls and Integrity Audit

At an intervention timestep, the selected membrane coordinate is placed above threshold before spike computation. Under the hard threshold rule in



Figure S1: S-MNIST loss-landscape diagnostic for SPRiF (top row) and LIF (bottom row). The 2D contours and 3D surfaces show cross-entropy over the same 25×25 perturbation-coordinate grid, with a shared logarithmic color scale and common vertical range. Each model uses its own checkpoint-normalized random directions, while evaluation samples and grid coordinates are matched.

Variant	Change relative to SPRiF full	Interpretation lock
$\omega = 0$	Removes rotational coupling while retaining the real decay path and slow-fast state separation.	Tests spectral rotation, not state separation.
Merged	Collapses the slow spectral state and fast discharge state into one three-dimensional state whose first coordinate is both membrane and reset target.	Tests the consequence of collapsing memory and discharge; it is not a pure reset-direction control and does not preserve the full model's total state dimensionality.
$\lambda = 0$	Fixes the reset direction to $[1, 0]^\top$, so reset subtracts only from the membrane coordinate.	Tests learned projective reset against scalar reset while preserving slow-fast separation.

Table S2: Mechanism variants corresponding to the main-paper ablation table.

Item	Setting
Input channels	30 total: 10 left-cue channels, 10 right-cue channels, and 10 noise channels, encoded as Poisson spike trains.
Pre-period	50 ms baseline period with noise channels active.
First / second cue	100 ms each; each cue is left or right.
Delay (train)	Drawn from $\{200, 400, 800, 1600\}$ ms.
Delay (eval)	Drawn from $\{200, 400, 800, 1600, 2500\}$ ms.
Target	Binary match/non-match label depending on whether the two cues are identical.
Intervention	At K random delay timesteps, selected membrane coordinates are placed above threshold before spike computation; each resulting spike follows the model's ordinary reset rule and recurrent pathway.
Intervention (train) fraction	15% of the 64 hidden units at each intervention timestep.
Intervention (eval) fraction	$\{5, 10, 15, 20, 30, 50\}\%$; the main-paper figure and table use the training-matched 15% slice.
K (train)	$\{0, 1, 2, 4, 8\}$.
K (eval)	$\{0, 1, 2, 4, 8, 16, 32, 40\}$; $K \in \{16, 32, 40\}$ are unseen during training.
Shared controls	Within each seed/delay/ K /fraction condition, models use matched batches, intervention masks, and evaluation probes.
Hidden units	64.
Architecture	One recurrent spiking hidden layer followed by a two-class non-spiking leaky readout. Previous hidden spikes enter the next hidden update through a learned 64×64 recurrent weight matrix.
Optimization	3,000 training steps; batch size 256; Adam with learning rate 3×10^{-3} ; gradient clipping at 1.0.
Rate regularization	Natural firing-rate target 0.10 with weight 0.05; intervention locations are excluded from the natural-rate term.
SPRiF variants	<code>sprif_full</code> , <code>sprif_lambda0</code> , and <code>sprif_merged</code> .
Baselines	LIF, ASRNN, and BRF with the same recurrent hidden width, optimizer, batch structure, and intervention masks.

Table S3: SI-DMS task construction, controlled intervention, and matched-control configuration.

Model	Clean	Acc ₈	Δ Acc ₈	Acc ₄₀	Δ Acc ₄₀
SPRiF full	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	99.8 $\pm$ 0.2	0.2 $\pm$ 0.2
SPRiF $\lambda=0$	99.4 $\pm$ 0.0	99.0 $\pm$ 0.1	0.4 $\pm$ 0.1	95.2 $\pm$ 0.2	4.2 $\pm$ 0.2
SPRiF merged	88.1 $\pm$ 1.4	55.8 $\pm$ 2.8	32.3 $\pm$ 1.4	51.7 $\pm$ 1.6	36.4 $\pm$ 2.7
ASRNN	100.0 $\pm$ 0.0	99.3 $\pm$ 0.5	0.6 $\pm$ 0.5	68.7 $\pm$ 7.4	31.2 $\pm$ 7.4
BRF	79.3 $\pm$ 1.4	70.7 $\pm$ 0.5	8.6 $\pm$ 1.4	53.7 $\pm$ 0.5	25.6 $\pm$ 1.2
LIF	74.9 $\pm$ 23.0	69.0 $\pm$ 26.0	5.9 $\pm$ 8.2	61.2 $\pm$ 19.2	13.7 $\pm$ 10.2

Table S4: SI-DMS accuracy and paired loss at the 15% intervention fraction (mean $\pm$ standard deviation over three seeds; percentage points). Each seed-level quantity is averaged over the five delays before cross-seed aggregation.

figures/si_dms_fraction_response.png

Figure S2: SI-DMS fraction response at the unseen intervention count $K = 40$. Curves are 3-seed means after averaging each seed over five delays; shaded bands show seed standard deviations. The point at 0% is the clean condition, and 15% is the intervention fraction used during training.

Eq. (??), this yields an actual spike and then applies the model’s ordinary reset rule. The intervention therefore does not bypass reset and does not add a separate post-spike update.

For SPRiF, let the reset direction be $\mathbf{r} = [1, \lambda_j]^\top$, so the fast-state reset identity is

$$\mathbf{u}_t - (\tilde{\mathbf{u}}_t - z_t \theta \mathbf{r}) = \mathbf{0}, \quad (4)$$

while the slow state has no reset update,

$$\Delta \mathbf{x}_t^{\text{reset}} = \mathbf{0}. \quad (5)$$

These are structural identities of the stated update. They formalize the SI-DMS distinction: forced spikes can alter the fast discharge state through real spike-and-reset events, but they do not directly modify the slow spectral state at that timestep.

4 Formal SPRiF Specification and Analysis Protocols

4.1 Slow Spectral State

The slow spectral state $\mathbf{x}_t \in \mathbb{R}^3$ evolves as

$$\mathbf{x}_t = A\mathbf{x}_{t-1} + B I_t, \quad (6)$$

where

$$A = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \rho \cos \omega & -\rho \sin \omega \\ 0 & \rho \sin \omega & \rho \cos \omega \end{bmatrix}, \quad B = \begin{bmatrix} 1 - \alpha \\ 1 - \rho \\ 0 \end{bmatrix}. \quad (7)$$

The first coordinate is a real exponential trace,

$$x_t^0 = \alpha x_{t-1}^0 + (1 - \alpha) I_t, \quad (8)$$

and the second and third coordinates form a damped rotational pair driven through the first oscillatory coordinate:

$$x_t^1 = \rho \cos \omega x_{t-1}^1 - \rho \sin \omega x_{t-1}^2 + (1 - \rho) I_t, \quad (9)$$

$$x_t^2 = \rho \sin \omega x_{t-1}^1 + \rho \cos \omega x_{t-1}^2. \quad (10)$$

The eigenvalues of A are α and $\rho e^{\pm i\omega}$. We parameterize

$$\alpha = \sigma(\alpha_{\text{raw}}), \quad \rho = \sigma(\rho_{\text{raw}}), \quad \omega = \pi \sigma(\omega_{\text{raw}}), \quad (11)$$

so $\alpha, \rho \in (0, 1)$ and $\omega \in (0, \pi)$, giving spectral radius $\max(\alpha, \rho) < 1$. These sigmoid parameterizations keep the slow spectral state strictly stable under the learned raw variables.

Bounded-input stability. Let $q = \max(\alpha, \rho) < 1$ and suppose $|I_t| \leq M$. Because the rotational block is ρR_ω , $\|A^k\|_2 = q^k$. Unrolling Eq. (??) gives

$$\|\mathbf{x}_t\|_2 \leq q^{t+1} \|\mathbf{x}_{-1}\|_2 + \|B\|_2 M \frac{1 - q^{t+1}}{1 - q}. \quad (12)$$

Thus the influence of the initial state decays geometrically and every bounded input sequence produces a bounded slow state. This guarantee concerns the intrinsic slow transition; it does not assert stability of an arbitrary learned network-level recurrent operator.

Under a constant input $I_t = c$, the real mode converges to c . The rotational pair converges to the unique fixed point

$$\mathbf{x}_*^{\text{osc}} = (I - \rho R_\omega)^{-1} \begin{bmatrix} (1 - \rho)c \\ 0 \end{bmatrix}, \quad (13)$$

$$R_\omega = \begin{bmatrix} \cos \omega & -\sin \omega \\ \sin \omega & \cos \omega \end{bmatrix}. \quad (14)$$

Thus the first oscillatory coordinate is not generally equal to c when $\omega \neq 0$.

For a unit input impulse at $t = 0$ with $\mathbf{x}_{-1} = \mathbf{0}$, the slow-state impulse response isolates the two retention modes separately:

$$x_k^0 = (1 - \alpha) \alpha^k, \quad (15)$$

$$x_k^1 = (1 - \rho) \rho^k \cos(k\omega), \quad (16)$$

$$x_k^2 = (1 - \rho) \rho^k \sin(k\omega), \quad (17)$$

for $k \geq 0$. The real mode provides monotone retention, whereas the rotational pair provides damped phase dynamics.

4.2 Fast Discharge State and Projective Reset

The fast discharge state $\mathbf{u}_t \in \mathbb{R}^2$ receives a projection of the slow spectral state through $G \in \mathbb{R}^{2 \times 3}$:

$$\tilde{u}_t^0 = \eta_0 u_{t-1}^0 + \kappa u_{t-1}^1 + (G \mathbf{x}_t)^0, \quad (18)$$

$$\tilde{u}_t^1 = \eta_1 u_{t-1}^1 + (G \mathbf{x}_t)^1, \quad (19)$$

with $\eta_0, \eta_1 \in (0, 1)$ parameterized as sigmoid-transformed raw variables. The membrane potential is $v_t = \tilde{u}_t^0$, and the spike is

$$z_t = H(v_t - \theta), \quad (20)$$

where H is the Heaviside step function.

Within each timestep, computation follows $I_t = W_{\text{in}} \mathbf{s}_t + W_{\text{rec}} \mathbf{z}_{t-1} + b \rightarrow \mathbf{x}_t = A\mathbf{x}_{t-1} + B I_t \rightarrow \tilde{\mathbf{u}}_t \rightarrow v_t = \tilde{u}_t^0 \rightarrow z_t = H(v_t - \theta) \rightarrow \mathbf{u}_t = \tilde{\mathbf{u}}_t - z_t \theta \mathbf{r}$. This

Audit item	Required behavior	Appendix status
Coverage	Each model/seed evaluates every configured delay-by- K -by-fraction cell.	All 18 runs contain 215 rows, giving 3,870 saved condition rows.
Forced-spike hit	Every selected unit should spike when $K > 0$.	The saved evaluation metrics record forced-hit rate = 1.0 for all 3,780 positive-intervention rows.
Shared controls	Comparisons should not depend on model-specific probes.	Within a seed/delay/ K /fraction condition, batches and intervention masks are shared across models.
Reset path	Forced spikes should trigger the same reset rule as ordinary spikes.	Interventions are applied pre-threshold; the resulting spike enters the standard reset update.
Recurrent path	Forced spikes should follow ordinary network communication.	The emitted spike is supplied to the learned recurrent pathway at the next timestep.
SPRiF slow-state insulation	A forced spike must not directly edit the slow spectral state.	Eq. (??) states $\Delta \mathbf{x}_t^{\text{reset}} = \mathbf{0}$; recurrent communication can only enter the next timestep through $W_{\text{rec}} \mathbf{z}_t$ when recurrence is present.

Table S5: SI-DMS integrity checks used to interpret the intervention results.

order matches the main paper and confines the local reset update to the fast discharge state.

Component-wise, projective reset is

$$u_t^0 = \tilde{u}_t^0 - z_t \theta, \quad (21)$$

$$u_t^1 = \tilde{u}_t^1 - z_t \theta \lambda_j. \quad (22)$$

Setting $\lambda_j = 0$ recovers scalar subtraction on the membrane coordinate; learned λ_j allows reset to act along $[1, \lambda_j]^\top$ in the fast state.

Direct-reset invariance. Denote the slow and fast states immediately before local reset by $(\mathbf{x}_t^-, \tilde{\mathbf{u}}_t)$ and immediately after reset by $(\mathbf{x}_t^+, \mathbf{u}_t)$. The update above gives

$$\mathbf{x}_t^+ = \mathbf{x}_t^-, \quad \mathbf{u}_t = \tilde{\mathbf{u}}_t - z_t \theta [1, \lambda_j]^\top. \quad (23)$$

Therefore the direct reset residual of the slow state is exactly zero for either spike outcome. This is a within-timestep structural guarantee, not a claim that future slow states are spike-independent: in a recurrent layer, z_t can affect I_{t+1} through the ordinary learned recurrent pathway.

4.3 Surrogate Gradient

The forward pass uses the hard threshold in Eq. (??). During backpropagation, the threshold derivative is

replaced by the multi-Gaussian surrogate ?:

$$\begin{aligned} \frac{\partial z_t}{\partial v_t} \approx & \gamma [a_0 \mathcal{N}(v_t - \theta; 0, \sigma_0^2) \\ & - a_1 \mathcal{N}(v_t - \theta; \mu_1, \sigma_1^2) \\ & - a_1 \mathcal{N}(v_t - \theta; -\mu_1, \sigma_1^2)], \end{aligned} \quad (24)$$

where $\sigma_0 = 0.5$, $a_0 = 1.15$, $\mu_1 = 0.5$, $\sigma_1 = 3.0$, $a_1 = 0.15$, and $\gamma = 0.5$. The forward dynamics remain unchanged.

4.4 Initialization, State Size, and Cost

The task-specific initialization ranges are listed in Table ???. For each decay $q \in \{\alpha, \rho, \eta_0, \eta_1\}$, a time constant τ_q in steps is mapped to $q = \exp(-1/\tau_q)$ before conversion to its unconstrained raw parameter. Raw variables are initialized so that the transformed α , ρ , η_i , and ω values fall in the listed ranges; after initialization, optimization proceeds in the unconstrained raw parameter space.

Each SPRiF neuron stores five state values $(x^0, x^1, x^2, u^0, u^1)$ and has 13 neuron-wise dynamics parameters: $\alpha_{\text{raw}}, \rho_{\text{raw}}, \omega_{\text{raw}}, \eta_{0,\text{raw}}, \eta_{1,\text{raw}}, \kappa$, the six entries of $G \in \mathbb{R}^{2 \times 3}$, and λ . The weight matrices W_{in} and W_{rec} and the bias b follow the layer architecture.

For a layer of width h , the local state update cost is $O(h)$ because each neuron uses fixed-size slow and fast states. When recurrent connectivity is enabled, the $O(h^2)$ multiplication $W_{\text{rec}} \mathbf{z}_{t-1}$ dominates for both LIF-style recurrent SNNs and SPRiF layers.

4.5 Analysis Protocols

For the impulse-response and temporal-kernel analyses in the main paper, we inject a unit impulse into trained SPRiF neurons and then record 100 steps of intrinsic slow-state evolution without further input. The four displayed SPRiF trajectories use the 10th, 40th, 70th, and 90th percentiles of learned α . For the GSC and QTDB cross-model panels, the 90th-percentile SPRiF real mode is paired with the available trained ASRNN membrane kernel having the closest derived τ_m , and each displayed kernel is normalized by its peak magnitude. Spectra from the real coordinate are aggregated across neurons. Because no subsequent input is applied and the slow state has no direct local reset term, the observed response shapes reflect the learned spectral parameters rather than later post-spike correction.

The learned reset-direction analysis uses the λ values shown in the main-paper mechanism figure. Across QTDB, GSC, and PS-MNIST, the reported λ distributions span both signs, and the main-paper figure reports weak Pearson correlations with firing rate, α , and ω ($|r| \leq 0.191$). This appendix treats those observations as evidence that projective reset directions remain diverse in the fast discharge state; it does not attach a universal biological meaning to either sign of λ .

4.6 Feature-Perturbation Diagnostic

Figure ?? reports a fixed-checkpoint feature-space perturbation diagnostic on GSC and QTDB. Controlled input corruptions are widely used to probe non-adversarial robustness ?, while additive Gaussian jittering is a standard magnitude-domain transformation for time-series inputs ?. Motivated by these evaluation practices, we apply additive Gaussian noise, random feature zeroing, and their combination to saved SPRiF and ASRNN checkpoints, with matched perturbation settings within each condition. This diagnostic is not a replacement for the five-seed benchmark table in the main paper and is not used as a statistical robustness ranking; it records how the saved checkpoints respond under the specified feature perturbations.



Figure S3: Feature-space perturbation diagnostic using saved checkpoints. SPRiF and ASRNN are evaluated on GSC and QTDB under additive Gaussian noise, random feature zeroing, and mixed perturbations. The clean values in this diagnostic come from the saved checkpoints used for the perturbation sweep and need not equal the five-seed benchmark summaries in the main paper.